

Quantitative fluorescence recovery after photobleaching (FRAP) is used to study the movement of molecules in live cells. During FRAP, the fluorescence of a green fluorescent protein (GFP)–tagged molecule is first measured and then photobleached in targeted cell regions. Researchers then assess the time course for fluorescence recovery, which is an indicator of molecular mobility of the GFP-tagged molecule into the photobleached region. For example, the average time required to recover 80% of the pre-bleach fluorescence of protein histone 1 (H1) fused to GFP (H1-GFP) in the photobleached region is given as t_{80} .

FRAP was used to analyze the mobility of H1 alone and in the presence of high-mobility group (HMG) proteins. Specifically, when H1-GFP is less able to bind chromatin, its mobility increases, allowing H1 to enter photobleached regions more rapidly, thus decreasing t_{80} . HMG proteins have been shown to have similar chromatin binding sites as H1 proteins. Figure 1 shows the chromatin binding motifs for different HMG proteins. Both H1 and HMG proteins can dynamically bind and unbind chromatin. Accordingly, researchers hypothesize that HMG competes with H1 for similar chromatin binding sites and therefore decreases H1 binding to chromatin.

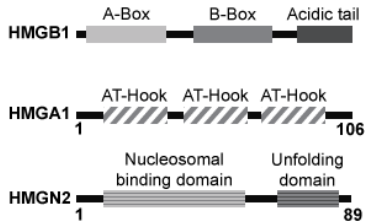


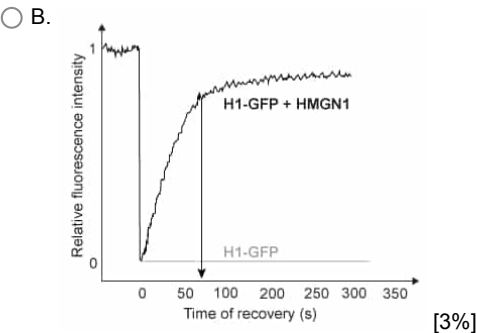
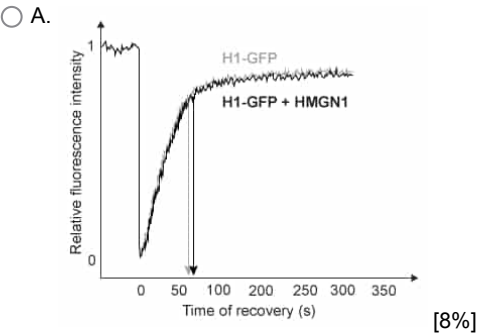
Figure 1 Unique chromatin binding motifs for different HMG proteins

During an experiment to test this hypothesis, either HMGB1 or HMGA1 proteins were microinjected into the cytoplasm of mouse embryonic fibroblast cells expressing H1-GFP, leading to abnormally high levels of these HMGs in the cells. Other fibroblasts received no injection. FRAP was performed on both euchromatin and heterochromatin in these cells. Euchromatin and heterochromatin domains were identified as regions weakly and strongly stained by H1-GFP, respectively. The relative intensity of H1-GFP fluorescence in chromatin of the cells was assessed, and t_{80} results are shown in Table 1.

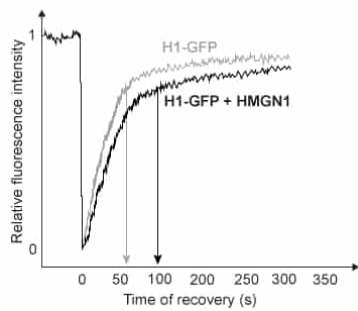
Table 1 Effect of HMGB1 and HMGA1 on H1-GFP Mobility

Experiment	Chromatin domain	t_{80} (s) Average time $\pm$ SD
H1-GFP	Euchromatin	104 $\pm$ 41
H1-GFP + HMGB1		46.0 $\pm$ 19
H1-GFP	Heterochromatin	126 $\pm$ 42
H1-GFP + HMGB1		82.3 $\pm$ 33
H1-GFP	Euchromatin	103 $\pm$ 31
H1-GFP + HMGA1		48.7 $\pm$ 40
H1-GFP	Heterochromatin	129 $\pm$ 40
H1-GFP + HMGA1		83.8 $\pm$ 21

Researchers identified another type of HMG protein, HMGN1, and hypothesized that HMGN1 also competes with H1 for chromatin binding sites. If researchers used the same experimental protocol described in the passage, which of the following FRAP analysis graphs would support this hypothesis? (Note: The t_{80} values for H1-GFP are indicated by the arrows.)

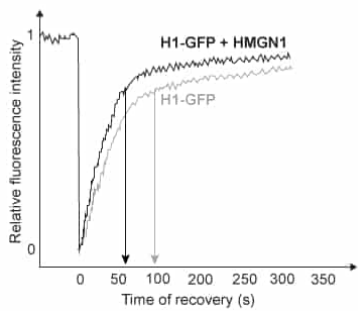


☐ C.



[32%]

☒ D.

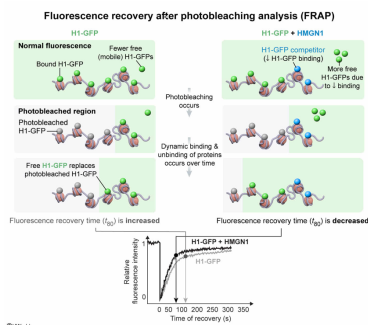


[56%]

Correct

56% answered correctly

Explanation:



Green fluorescent protein (GFP) is frequently used to track the mobility of proteins in a cell. The passage describes quantitative fluorescence recovery after photobleaching (FRAP), a process in which the fluorescence of a GFP-tagged molecule is first measured and then destroyed in targeted cell regions. The time course for fluorescence recovery in the photobleached region can then be assessed as a measure of molecular mobility for the GFP-tagged molecule. Specifically, t_{80} is the average amount of time required to recover 80% of the pre-bleach fluorescence.

In addition, the passage states that when H1-GFP is less able to bind chromatin (eg, in the presence of a competitor), the mobility of H1-GFP *increases*, allowing it to enter photobleached regions more rapidly during FRAP, which *decreases* t_{80} . Table 1 shows that in both heterochromatin and euchromatin, the presence of an H1 competitor (ie, different HMG proteins) *decreases* t_{80} as expected.

In this question, researchers identified another HMG protein (HMGN1) and hypothesized that **HMGN1** also **competes with H1** for the **same chromatin binding sites**. If this is true, the addition of HMGN1 should *increase* the mobility of H1-GFP, consequently *decreasing* t_{80} for H1-GFP during FRAP. To support this hypothesis, the FRAP analysis graph should show that **H1-GFP + HMGN1** has a **lesser (not greater) t_{80} value** than **H1-GFP alone (Choice C)**. Only the graph in **Choice D** shows this result.

(Choice A) Because this graph shows almost no difference in the t_{80} values of both states, this data indicates that HMGN1 likely does *not* compete with H1 for chromatin binding sites, which does *not* provide support for the hypothesis.

(Choice B) This graph shows that H1-GFP never recovered fluorescence after photobleaching. However, dynamic H1 binding and unbinding to chromatin would allow eventual fluorescence recovery.

Educational objective:

Green fluorescent protein (GFP) is frequently used to track the mobility of proteins in a cell. The protein of interest is tagged with GFP, and its movement, localization, and expression can be studied in cells based on its fluorescence.

Time spent:0 sec
Subject:
Biology

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System:
Molecular Biology